

Ayushmaan Das

AI/ML Engineer & Software Developer
LLMs · RAG · Agentic AI · Edge AI · MLOps

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Education

- **Sri Ramachandra Faculty of Engineering and Technology, SRIHER** *2021 - 2025*
  B.Tech Computer Science and Engineering (Artificial Intelligence and Machine Learning) CGPA: 9.91

Experience

- **MulticoreWare Inc.** *Chennai (On-site)*
  Software Engineer July 2025 - Present
  - Developed an embedded-systems IDE to build, run, and flash firmware on customer-specific boards, with an AI Pipeline for import, pre-check, calibration, and quantization; Model Visualizer, Graphs, Memory Estimator, Model Zoo, Latency Estimator, Debug Dashboards, Power Consumption, AI Embedding, and SDK code generation tools
  - Built AI Playground (RAG: embedding + re-ranking, Fine-tuning, multi-tenant Adapter Inference); live in production on AWS (S3, EKS)/Kubernetes with Prometheus, LLM-token pod autoscaling, Stripe billing

  Junior Engineer *January 2025 - June 2025*
  - Enabled coding-assistant LLMs (CodeLlama-7B/13B, Codestral-22B, Qwen2.5-Coder-32B, Mxfp6-quantized) to run on Edge AI based Qualcomm hardware (C++); built async Inference Server with Load Balancing, improved performance by supporting KV Caching (results up to 33.71 tok/s), LiteRT, AIMET quantization, MLFlow experiments

- **Upraised (JobStop Inc.)** *June 2024 - December 2024*
  Software Development Intern (ML Team) Remote
  - Built a RAG-powered backend to auto-generate assessments from prompts, and an AI proctoring system with Pyannote speaker diarization flagging anomalous audio; used Transformers, Hugging Face, Azure DevOps

- **RecruitNXT** *February 2023 - November 2023*
  Python Developer and ML Intern Remote
  - Designed a resume-parsing pipeline from scratch, building and annotating datasets to train SpaCy-based NER models for candidate data extraction, and containerized it with Docker for production deployment

Personal Projects

- **Medical Agentic RAG** *2026*
  Dual-pipeline diagnostic assistant for Medical-field related Q/A: fixed RAG vs. LangChain ReAct agentic orchestration
  - LangChain ReAct agent (Llama 3.3 70B), FAISS-based RAG with fine-tuned FLAN-T5, Dockerized BERT classifier

- **Spotify Music Recommendation Engine** *2025*
  Emotion-based (using CNN) and custom ML algorithm-based song recommendation web application integrated with Spotify
  - Custom CNN model from scratch using TensorFlow and Keras; Cosine Similarity, Streamlit, Spotify API (playlists)

Skills and Interests

Languages: Python, C++, C, Java, SQL, NoSQL
AI/ML Tools: LangChain, Agentic AI, ReAct Agents, MCP (Model Context Protocol), KV Caching, RAG, Fine-Tuning, TensorFlow, Keras, PyTorch, Scikit-Learn, OpenCV, Hugging Face, SpaCy, FAISS, Sentence-Transformers, LiteRT, Quantization, AIMET, MLFlow, Neural Networks, Deep Learning, Reinforcement Learning, MLOps, Prompt Engineering
Frameworks: Flask, FastAPI, Streamlit, Gradio
Cloud & Infra: AWS (S3, EKS), Azure DevOps, OCI, Kubernetes, Docker, Prometheus, Koyeb, CI/CD
Other Tools: Power BI, Git, GitHub, Linux, MongoDB, APIs, PlatformIO, IDE Development
Certifications: Docker (Great Learning), Claude Code 101 & Introduction to MCPs (Anthropic)
Soft Skills: Communication, Problem Solving, Critical Thinking, Time Management, Leadership, Creativity
Linguistics: English (Conversational, Read & Write), Hindi & Bengali (Native, Bilingual)

Achievements

- **Book Chapter** in Composite AI: Fundamentals, Challenges, and Applications (Chapter 10) *2026*
- **Spot Award** (MulticoreWare Inc.) for outstanding delivery, ownership and reliability *2026*
- **Founder Chancellor Gold Medal (SRIHER)** for Best Outgoing Student *2025*
- **Book Chapter** in Artificial Intelligence and Machine Learning for Smart Community *2024*
- **Book Chapter** in Advancement of Data Processing Methods for Artificial & Computing Intelligence *2024*